

Tanish Arora

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LinkedIn

GitHub

Portfolio

EDUCATION

Vellore Institute of Technology, Chennai
B.Tech in Computer Science Engineering

Aug 2023 – Expected June 2027
CGPA: 8.79

TECHNICAL SKILLS

- **Languages:** C++, JavaScript, TypeScript, SQL, Bash/Shell
- **Frameworks & Libraries:** React, Node.js, Express, FastAPI, Tailwind, REST APIs
- **Databases & Cache:** PostgreSQL, Redis, MongoDB, SQLite
- **Data & Machine Learning:** FAISS, Scikit-learn, XGBoost, ARIMA, Pandas, NumPy
- **DevOps & Architecture:** AWS (EC2, ECS, ECR, ALB), Docker, Celery, GitHub Actions, Linux
- **Developer Tools:** Git, GitHub, VS Code, Postman, Jupyter Lab

EXPERIENCE

HCLTech

May 2026 – Jul 2026

AI & GenAI Research Intern

Remote

- Contributed to AI-driven **EEG signal analysis** projects, leveraging **Transformer architectures** and **Kolmogorov-Arnold Networks (KANs)** for neural time-series modeling.
- Developed data preprocessing pipelines and performed feature extraction to improve model classification performance.
- Conducted experiments evaluating deep learning models, gaining hands-on experience in biomedical signal processing and research-oriented AI development on real-world neuroscience applications.

Sahii.in

Apr 2026 – May 2026

Software Developer Intern

Remote

- Architected and optimized full-stack web applications using **React**, **Node.js**, and **FastAPI**, implementing technical SEO enhancements that drove a **150% increase in user traffic**.
- Slashed application loading time by a full second—achieving an ultra-low **15ms latency**—by restructuring and refining polyglot persistence data architectures using **MongoDB**.
- Streamlined cloud infrastructure workflows by containerizing applications with **Docker** and seamlessly migrating large-scale organizational datasets to **AWS (ECS/EC2)** and **Cloudflare R2**.
- Independently resolved complex architectural challenges while collaborating effectively within a fast-paced team environment.

PROJECTS

Commons – Real-Time AI Moderation Gateway [\[github\]](#) [\[demo\]](#)

Feb 2026

React, FastAPI, Node.js, Redis, FAISS, Celery, PostgreSQL, Sentence-Transformers

- Architected a sub-second moderation pipeline using **FastAPI** and **Node.js** for concurrent, multilingual text and voice streams.
- Engineered a **Redis** pre-filter to intercept ~70% of invalid traffic in <10ms, drastically reducing LLM inference costs.
- Integrated **FAISS** for <20ms semantic search on banned embeddings, cascading edge cases to a Groq LLM to prevent hallucinations.
- Designed an asynchronous architecture by offloading audit logging to **PostgreSQL** via **Celery** workers for instant client feedback.

NeuroSense – AI-Powered Dementia Screening Platform [\[github\]](#)

Mar 2026

React 19, Node.js, FastAPI, Vosk, Scikit-learn, OAuth, Microservices

- Architected a modular 3-tier microservice platform merging classical cognitive test metadata with acoustic speech feature extraction to compute probabilistic dementia risk scores.
- Trained a **scikit-learn** model on 1,600 clinical samples, achieving **81.5% accuracy** and an **89.5% ROC-AUC** for dementia risk classification.
- Engineered a **Node.js** API gateway with **OAuth** authentication, automated PDF reporting, strict proxy timeout controls, and latency tracing via a dedicated ScreeningAuditLog for fault-tolerant observability.
- Built a Vite-powered **React 19** dashboard (~31KB critical bundle) and integrated an offline **Vosk** engine achieving 0.15s transcription latency for real-time, privacy-centric inference.

Secure Plus – Privacy-Focused Document Redaction Tool [\[github\]](#)

Dec 2025

Python, OpenCV, Node.js, AWS ECS, Docker, MERN Stack

- Developed a **Python and OpenCV** engine to accurately detect and redact sensitive PII fields (PAN, emails, DOB) from documents.
- Implemented irreversible **Gaussian blur masking** to secure user-selected regions while preserving the original document layout.
- Deployed heavy image-processing tasks to containerized Python workers on **AWS ECS** to ensure scalable architectural performance.